



Bhargav
Sr. Cloud / DevOps Engineer
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PROFESSIONAL SUMMARY:

Around **6 years** of professional experience in IT Industry which includes System Administration working on Red Hat Enterprise Linux (RHEL), Cent OS, Ubuntu, Sun Solaris, and windows environment, Amazon Web Services, Microsoft Azure, DevOps as well as Build and Release Engineer, Continuous Integration/Continuous Deployment, supporting development, Deployment operations in different environments.

- Expertise in all phases of **Software Development Life Cycle (SDLC)**, including **Analysis, Design, Development** and **Testing** of Client - Server and Web-based n-tier Architecture for **web applications** with exposure to diverse business domains.
- Hands-on experience with AWS and Microsoft Azure Cloud services, Storage Accounts and Virtual Networks and provisioning of **IAAS & PAAS Virtual Machines** and Web\Worker roles on Microsoft Azure Classic and Azure Resource Manager.
- Expertise in working with **AWS** provided solutions like **EC2** and **ELB** which includes creating EC2 instances, adding EBS volumes for EC2 and familiar with **VPC, Route 53, S3, IAM, RDS**.
- Experience in IaC using PowerShell – **ARM, AZ cli, Bicep, terraform** to improve reutilization of modules across and easy to maintain. Developed modules for **Azure, AWS** IaaS, PaaS, DBaaS services, worked with **Terraform** key features such as Execution plans, Resource Graphs and Change Automation.
- Experience in infrastructure migrating an On - premises Instances or **Azure Classic Instances** to Azure **ARM Subscription** with Azure Site Recovery.
- Well Experienced in App Containerization using Docker for Tightly coupled (Docker Compose) and Micro services (Docker) and Deployed in hybrid Kubernetes cloud environments. Used packaging tools, helm and **Argo CD**, maintained different release cycles for production and non-production. Enabled CI & CD using **Jenkins** on **Linux, Azure DevOps, TeamCity** project specific tools. Used Docker Registry, ACR and ECR to maintain the image versions. And for build: msbuild, ant, dotnet, npm, yarn.
- **Ansible and AWS lambda, elastic cache, and cloud watch** logs to automate the creation of log aggregation pipeline with ElasticSearch, Log stash, Kibana stack (ELK stack) and expertise in deploying the code to **AWS servers** through the tool **Ansible** and have well hands on writing it's **yaml**.
- Worked with **Ansible Centralized Server** (Tower) and created the playbooks to support various middleware application servers and was involved in configuring the Ansible tower as a **configuration management tool** to automate repetitive tasks.
- Experience in compiling and deploying the different applications (Apache, Tomcat, windows etc.) on different Linux and windows servers through DevOps pipeline which includes **Chef, Jenkins, Artifactory and RedHat OpenStack**. Extensive knowledge in continuous integration tool Jenkins with different plugins like **GitHub, Artifactory, etc**.
- Having experience in Agile Methodologies, Scrum and sprints experience, along with **data analytics, data wrangling and Excel data extracts**.
- Having experience in creation, managing **Grafana Dashboards** for **monitoring the Infra: Instances, Kubernetes cluster** using **Prometheus** for gauging the Availability of the API server, scheduling rate of the Scheduler and using PromQL queries for specific metrics. Written **Ansible** Script for **ELK package** to be installed, deployed, and configured in AWS cloud environment and build CI/CD Pipeline with different Jenkins jobs running.
- Experience working on **New Relic** to respond to incidents, diagnose problems and minimize downtime to ensure seamless deployments of DevOps pipelines.
- Experience working on User admin groups, maintaining account and monitoring system performance using **Nagios** and **Splunk**.
- Configuration and troubleshooting across a variety of platforms and deployed new Splunk instances, including **clustered deployments and integrated Splunk** with a wide variety of legacy data sources and industry leading **commercial security tools**.

TECHNICAL SKILLS:

Cloud Services	AWS, Azure
AWS Services	EC2, VPC, RDS, AMI, IAM, Cloud Formation, S3, Cloud Trail, SNS, CLI, Route53, Auto Scaling, AWS Lambda, DynamoDB, Redshift, Database Migration Service, Cloud Watch.
Microsoft Azure	Networking (VPN, Load Balancer, Firewalls), Azure Site Recovery, Azure Backup, Azure API, Azure Services, Azure Security Center, Azure Monitoring.
Configuration Management Tools	Ansible, Chef, Puppet
Container Management Tools	Docker, Kubernetes: (Rancher with Azure, AKS and EKS), OpenShift, Docker swarm.
Operating Systems	Windows, Linux, Unix, Mac OS X.
Automation Tools	ANT, Maven, Jenkins, Terraform
Scripting languages	Python, Ruby, Bash shell, Power shell, JSON
Version Control Tools	GIT, Bit bucket, GITLAB, GITHUB, Azure DevOps
Monitoring Tools	Splunk, Nagios, CloudWatch, ELK. Azure Monitoring, App Insights, New Relic

PROFESSIONAL EXPERIENCE:

Role: DevOps Engineer/ Release Manager **Nov 2021 -Present**
Client: BuildingLink, NewYork City, NYC

Responsibilities:

- Designing, Implementing, and Maintaining all Azure infrastructure and services within a managed service environment. Migrating from On-Premises Infrastructure to Azure Cloud and configured various Azure Services including **VMs, RDS, Networking & Security, Policies, Storage: Blobs, Queues, Tables, App Insights, Azure Monitor, Load Balance Route Table, SQL, Data Bricks, Data Factory, ACR, and AKS** etc.
- Design, Deploy and maintain enterprise class security, network, and systems management applications within an **Azure Environment**. Created design documents with the Azure Security best practices to be implemented within the team.
- Implemented best practices for securing Java micro services on AKS, including network policies, **role-based access control (RBAC)**, and secrets management. Worked on deploying Java micro services on **Azure Kubernetes Service (AKS)** using deployment pipelines maintaining and configuring CI/CD pipelines for Java micro services deployments on to **AKS**.
- Integrated **Azure Container Registry** and **AKS**, providing a comprehensive ecosystem for application development and management by using **Service Fabric**.
- Worked on automation of scripts to configure and manage **AppDynamics** settings, including application, server configurations.
- Worked on **Rancher Kubernetes with Azure** for a **Vue.js** application and integrated with **Hashicorp Enterprise Vault** to maintain Secrets in the **Micro Services**.
- Worked on defining **Azure infrastructure resources**, such as **virtual machines, virtual networks, storage accounts, and load balancers, using Terraform modules to encapsulate and reuse infrastructure configurations across multiple Azure deployments**.

- Designed **Azure CLI** to create custom sized VPC, subnets, NAT to deploy Web applications & database Services. Designed roles and groups for users & resources using **Active Directory and Replications**.
- Used **TeamCity** pipelines to drive all micro services builds out to the **NuGet registry** and then deployed to Kubernetes, Created Pods and managed using **Kubectl**.
- Maintained well release cycles, used **Blue Green** deployment and **F5 Load Balancer** to flip the agent pools.
- Virtualized servers in Docker as per test environments and Dev-environments requirements and configured automation using **Docker containers**. To achieve Continuous Delivery goal on high scalable environment, used **Docker** coupled with load-balancing tool **Nginx**.
- Involved in creating or updating custom dashboards in **AppDynamics** by using **Powershell scripts**.
- Used **Carbon Black** to scan the artifacts in pipelines as part of quality check in Azure DevOps, used Test Driven Development – **TDD** by integrating **NUNIT** test cases check in the pipelines, also we have **Sonar Cloud** to ensure Code Quality. Worked on **IOS-XCode** and **Identity Server (IS)** upgradation.
- Used **PINGDOM** for Site availability and integrated with DEVOPS Teams groups to maintain Zero Downtime or Alerting.
- Integrated all our Applications, Micro services and Hosts with **New Relic** to have monitoring and alerting.
- Used **yarn** to package **Vue.js** applications and to know vulnerabilities and packing the applications.

Environment: Azure, Azure DevOps, TeamCity, Hashicorp Vault - Enterprise, Docker, Rancher Kubernetes with Azure, PowerShell, Terraform, Sonar Cloud, F5 Load Balancer, GitHub, JIRA, Confluence, Carbon Black, New Relic.

Role: Site Reliability Engineer / OpenShift Admin **Sep 2020 – Nov2021**

Client : Brother International Corporation, Bridgewater City, NJ

Responsibilities:

- Designing, Implementing, and maintaining all AWS Infrastructure and services within a managed service environment. Migrating from On-Premises Infrastructure to AWS Cloud and configured various **AWS** Services including **EC2, RDS, VPC, S3, Glacier, CloudWatch, CloudFront, Route53, SQS, SNS** etc.
- Design, Deploy and maintain enterprise class security, network, and systems management applications within an **AWS Environment**. Created design documents with the AWS Security best practices to be implemented within the team.
- Designed **AWS** Cloud Formation templates to create custom sized VPC, subnets, NAT to deploy Web applications & database templates. Designed roles and groups for users & resources using **AWS Identity Access Management (IAM)**.
- Expertise in integrating **Terraform** with Ansible, Packer to create and Version the AWS Infrastructure, designing, automating, implementing and sustainment of Amazon machine images (AMI) across the **AWS Cloud environment** and worked on Data Centre migration of applications from Data Centre to AWS Cloud using lift and shift strategy.
- Extensively involved in infrastructure as code, execution plans, resource graph and change automation using **Terraform**.
- Designed **Terraform** templates to create custom sized **VPC, subnets, NAT** to ensure successful deployment of **Web applications** and database templates on AWS and worked on creating the modules driven AWS Infrastructure with **Terraform** and created Infrastructure **Git** repositories for **Terraform** to launch the stacks.
- Expert in writing Templates for **AWS** infrastructure, as a code using **Terraform** to build Development, Testing, staging and production environments. Worked extensively with **Terraform** and wrapped it **Terragrunt** and managed the back-end state in **S3 Bucket**.
- Created E1 and E2 Test environments and worked on **OpenShift** platform in managing Docker containers and Kubernetes Clusters and Created Kubernetes clusters using **Ansible** playbooks.
- Worked as an **OpenShift** SMEs for migration of various applications from **OpenShift 2.x** to the newer 3.x platform for some major applications already existing in the V2 environment.

- Working knowledge of Docker Hub, Docker Container network, creating Image files primarily for middleware installations & domain configurations. Evaluated **Kubernetes** for Docker container **Orchestration**.
- Created Clusters using **Kubernetes** and worked on creating many pods, replication controllers, replica sets, services, deployments, labels, health checks and ingress by writing **Yaml** files.
- Used **Jenkins** and pipelines to drive all micro services builds out to the **Docker-registry** and then deployed to Kubernetes, Created Pods and managed using **Kubernetes**.
- Virtualized servers in Docker as per test environments and Dev-environments requirements and configured automation using **Docker** containers. To achieve Continuous Delivery goal on high scalable environment, used **Docker** coupled with load-balancing tool **Nginx**.
- Written many **Ansible Playbooks** with modules, tasks, roles, facts, and templates for VAR files, also configured files by conditions by keeping dynamic values.
- Triggered YML files in Ansible ran **Ansible** playbooks and created various roles for applications, then deployed the Applications/Services on hosts.
- Developed a fully automated continuous integration system using Git, MySQL, Jenkins, and custom tools developed in **Python**.
- Automated AWS S3 data upload / download using Python scripts and wrote Python scripts for pushing data from DynamoDB to MySQL Database and automated **JIRA** processes using Python and bash scripts.
- Experience executing the CI Jenkins build job for both Android and iOS application builds. Using **GIT** tool as the source code repositories for all projects and **Artifactory** for all builds release repository.

Environment: AWS EC2, S3, VPC, Route 53, CloudFormation, CloudWatch, OpenShift, Docker, Kubernetes, Ansible, Terraform, Maven, IAM, Jenkins, Chef, GIT, JIRA, Nagios, Splunk, RHEL, CentOS, Shell scripting, Python Scripting, Splunk.

Role : Cloud/DevOps Engineer

Febr 2020 – Sep 2020

Client: Chase, Wilmington, DE

Responsibilities:

- Created recommendations on how to duplicate a subset of on-premises machines to the **Azure Infrastructure as a Service (IAAS)** offering which will be used for disaster recovery. This analysis includes the specifics to synchronize on-premises data with SQL Server and SharePoint instances hosted in VMs. **Azure templates** for quick deployments and advanced **PowerShell** scripting.
- Worked on Azure Site Recovery and Azure Backup- Deployed Instances on Azure environments and in Data centers and migrating to Azure using **Azure Site Recovery** and collecting data from all Azure Resources using Log Analytics and analyzed the data to resolve issues.
- Worked on Server less services, created and configured **HTTP** Triggers in the Azure Functions with application insights for monitoring and performing load testing on the applications using the Visual Studio Team Services (VSTS) also called as **Azure DevOps Services**.
- Playing key role in migrating the existing on-premises applications to **AWS VPC**. Creating a **Lambda** Deployment function and configured it to receive events from your S3 bucket. Developing API for using **AWS Lambda** to manage the servers and run the code in the **AWS**.
- Managed AWS infrastructure as code using **Terraform** and used Terraform scripts to Automate Instances for Manual Instances that were launched before. Written Templates for **Azure Infrastructure as code** using **Terraform** to build staging and production environments.
- Used **Kubernetes** to deploy scale, load balance and manage Docker containers with multiple name spaced versions and experienced deploying container-based systems such as Docker or Kubernetes on private and public clouds (OpenStack or AWS).
- Building/Maintaining **Docker** container clusters managed by **Kubernetes Linux, Bash, GIT, Docker** and configured the Weave Net such that it creates the Virtual networks which connects Docker Containers to multiple hosts.
- Managed **Kubernetes** charts using Helm, Created reproducible builds of the Kubernetes applications, managed Kubernetes manifest files and Managed releases of Helm packages.
- Hands-on experience in implementing **Ansible** and Ansible Tower as Configuration management tool, to automate repetitive tasks, quickly deploys critical applications.

- Proactively manage change by writing Python code by using **Ansible** Python API to automate Cloud Deployment Process.
- Implemented a continuous Delivery pipeline with **Docker, Jenkins, and GitHub**. Responsible for installation & configuration of **Jenkins** to support various **Java builds**.
- Used Jenkins plugins to automate continuous builds and publishing Docker Images to the **Nexus Repository**.
- Built custom tools in **python** for generating email templates which are powerful enough to consume large amount of data and convey the testing results in a simpler way.
- Worked on **JIRA** for tracking and updating the JIRA tickets and per assigned and installed various Jira plugins such as Jira client, Jira importer plugin, Jira Charting Plugin, the connector for Microsoft project and Jira Misc. Custom fields.
- Built **Elastic search, Log stash and Kibana (ELK)** for centralized logging and then store logs and metrics into **S3** bucket using **Lambda** function.
- Used **Ansible** to Setup/teardown of **ELK** stack and troubleshoot the build issues with **ELK** and work towards the solution.

Environment: Azure, AWS, PowerShell Scripting, Docker, Kubernetes, Python, Bash, Linux, Maven, Jenkins, Terraform, Ansible, JIRA, ELK, Python, Shell scripts, Splunk, GIT, ADFS.

Role: Software Development Engineer

Sep 2018 – Dec 2019

Client: Covent Informatics , India

Responsibilities:

- Worked on Amazon Web Services (**AWS**) for a multitude of applications utilizing the AWS stack such as EC2, VPC, Glacier, Route53, S3, RDS, Cloud Watch, Cloud Trail, SNS and IAM, focusing on high-availability, fault tolerance, Load balancing and auto-scaling in designing, Deploying, and configuring.
- Successfully migrated the technology stack from a data center with failing hardware and frequent service interruptions to a secure and highly available hybrid cloud infrastructure on **AWS** Web Services.
- Created an **AWS** RDS Aurora DB cluster and connected to the database through an Amazon RDS Aurora DB Instance using the Amazon RDS Console.
- Create and maintain highly scalable and **fault tolerant** multi-tier AWS and Azure environments spanning across multiple availability zones using **Terraform** and **CloudFormation**.
- Written Templates for Azure Infrastructure as code using **Terraform** to build staging and production environments.
- Integrated Azure Log Analytics with Azure VMs for monitoring the **log files**, store them and track metrics and used **Terraform** as a tool.
- Worked on **OpenShift** for **container orchestration** with **Kubernetes** container storage, automation to enhance **container** platform multi-tenancy.
- Expertise in **Chef** orchestration, Install and configure system components such as FTP, DB2, File server, build server, Batch server, WebLogic etc. using chef automation from scratch.
- Written **Chef Cookbooks**, recipes using Ruby to automate the installation of Middleware Infrastructure like Apache Tomcat, JDK, and configuration tasks for new environments.
- Configured **Jenkins masters and slaves** on multiple machines and administer all the instances. Integrated Stash/Bitbucket with Jenkins to start automated builds on Jenkins using various triggers such as pull request, tags and commits.
- Built scripts using **MAVEN** build tools in Jenkins, to deploy J2EE applications to Application servers from one environment to other environments.
- Implemented zero downtime deployment process in WebLogic using python, and shell script and automated it using Jenkins.
- Used pytest framework to integrate several **python** tools for testing applications which involves xdist, mock, parallel, selenium, chrome, Firefox and provide html/xml reports.
- Coordinate/assist developers with establishing and applying appropriate branching, labeling/naming conventions using **GIT** source control. Imported and managed multiple corporate applications into GitHub code management repositories.

- Installing and configuring tools like **Splunk** for application servers, Integration with Splunk, which offers a pre-built knowledge base of critical dashboards and reports and created **Splunk** Search Processing Language (**SPL**) queries, Reports and Dashboards.

Environment: AWS, EC2, VPC, S3, RDS, IAM, CloudWatch, CloudTrail, Chef, Apache Tomcat, Jenkins, Maven, Shell Scripts, GIT, Splunk, Python, Kubernetes, JDK.

Education Details: University of Dayton - Masters in computer Science